

Combustion reactions

Learning outcomes

By the end of this section you should be able to deduce equations for reactions of combustion, including hydrocarbons and alcohols.

‘Fire has long been perceived as one of, if not the key, human technological innovations, underlying human evolutionary success, including our widespread distribution...’

Katharine MacDonald (2017) 
(<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5489006/>)

Without fire, where would we be as a species? Its discovery has given us a way to cook and preserve our food, not to mention providing us with warmth and light. At the root of fire is a chemical reaction known as combustion. Yet, what is combustion and what types of materials can be combusted?



Figure 1. What would have become of humans without the ability to harness combustion reactions?

Credit: Maria Korneeva, Getty Images

A combustion reaction involves two important reactants:

- fuel (a combustible substance)
- oxygen gas.

In all cases, the combustible material reacts with oxygen, and produces metallic or non-metallic oxides. These reactions are highly exothermic since the molecules produced are energetically more stable than the reactant fuel and oxygen gas.

Fuels can fall into a variety of categories, from reactive metals and non-metals to organic molecules like hydrocarbons and alcohol. The majority of organic molecules undergo combustion reactions.

Combustion of metals

Nearly all metals react with oxygen to form metallic oxides. Recall from sections S3.1.4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/group-1-and-17-elements-id-44694/>) and S3.1.5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-and-nonmetallic-continuum-id-44695/>), that the most reactive of these metals are the alkali metals followed by the alkaline earth metals. When reactive metals combine with oxygen they can produce tremendous amounts of energy, both in the form of heat and light. This type of reaction is shown below, in the reaction of magnesium metal with oxygen gas.



Burning magnesium ribbon



Video 1. The combustion of magnesium.

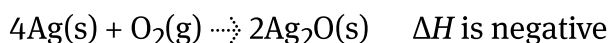
 More information for video 1

The interactive video demonstrates the combustion of magnesium ribbon. A strip of magnesium ribbon is clamped in place with a metal stand and ignited using a matchstick flame.

Initially, magnesium is a shiny, silver-grey metal. When heat is applied to one end of the ribbon, the metal reaches its ignition temperature and begins to burn in the presence of atmospheric oxygen.

Upon ignition, the magnesium burns with an extremely bright white flame. As the reaction progresses, it produces a white, powdery ash and a stream of white smoke. This white ash is magnesium oxide (MgO), the product of the reaction between magnesium and oxygen. The aftermath of the reaction leaves behind a noticeable transformation. The once-solid, metallic ribbon has disintegrated into a powdery white substance hanging loosely from the clamp.

Less reactive metals also combine with oxygen. For example, iron, copper, and silver react with oxygen to form their own metallic oxides. We often describe these chemical reactions as rusting or tarnishing. Although the equation showing the reaction of silver and oxygen might look similar to a combustion reaction, the reactions of oxygen with less reactive metals occur at a much slower rate.





Credit: Brian Caissie, Getty Images



Credit: bridixon, Getty Images



Source: "[Shugar-milk-hg](https://commons.wikimedia.org/wiki/File:Shugar-milk-hg.jpg)  (<https://commons.wikimedia.org/wiki/File:Shugar-milk-hg.jpg>)" by

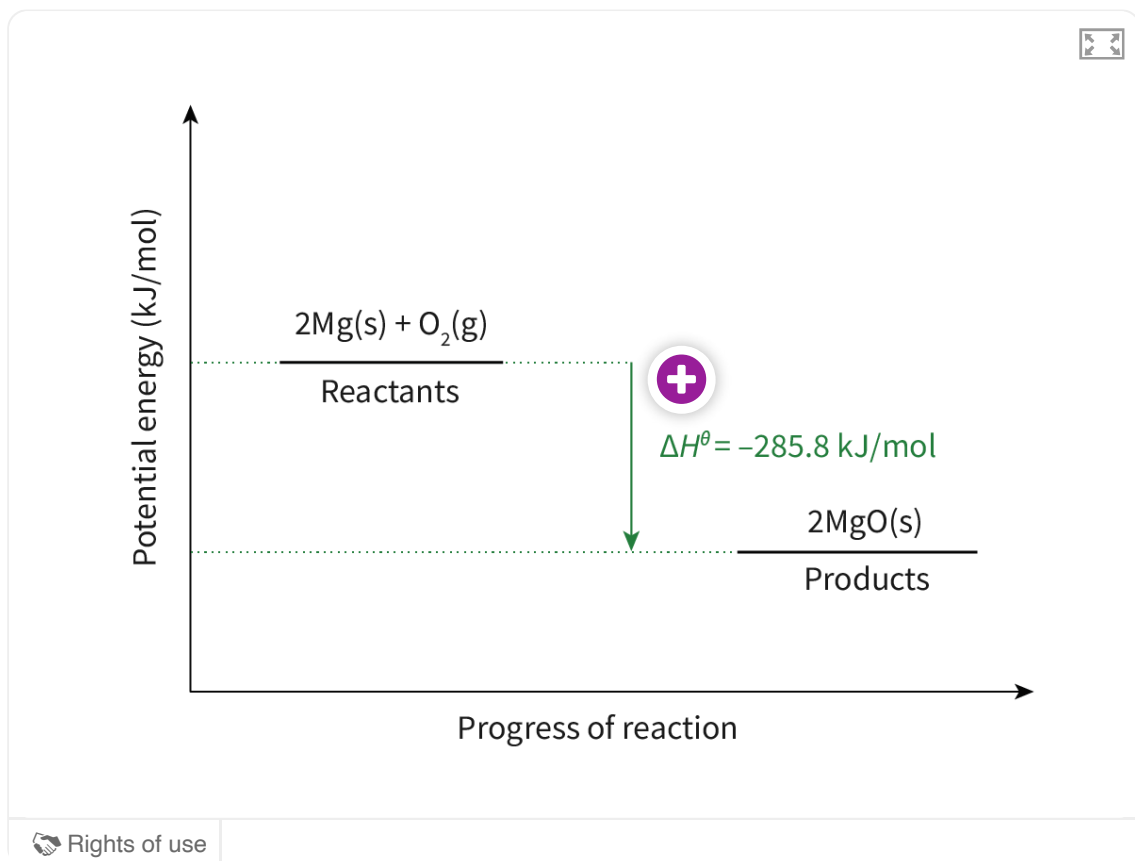
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Figure 2. Metal and oxygen reactions.

Making connections

As described in subtopics R1.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43468/>) and R1.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43471/>), energy is supplied into the system to break the bonds between the reactant atoms. More energy is released to form the bonds within the product molecules than is absorbed to break the bonds in the reactant molecules. As the product molecules form, and become more energetically stable, energy is released from the system to the surroundings.



Interactive 1. Enthalpy Diagram for the Combustion of Magnesium to Form Magnesium Oxide.

More information for interactive 1

This interactive energy graph illustrates the enthalpy change (ΔH°) during the combustion of magnesium (Mg) to form magnesium oxide (MgO). It visually represents how potential energy changes as the reaction progresses from reactants to products. The graph x-axis represents the progress of the reaction, showing the transition from reactants to products. The y-axis represents the potential energy (kJ/mol). The Reactants: At the beginning of the reaction, magnesium (Mg) and oxygen (O₂) have high energy [2Mg (solid) + O₂ (gas)]. The downward green arrow represents the energy released during the reaction. The enthalpy change (ΔH°) is -285.8 kJ/mol , indicating that energy is lost to the surroundings.

The hotspot represented by a plus sign is given beside the green arrow. By clicking the

hotspot, it reveals the image of burning magnesium reaction. The image illustrates a bright white flame produced when magnesium metal burns in oxygen. A person is holding magnesium with tongs. The reaction results in the emission of white light and smoke, indicating the formation of magnesium oxide (MgO).

The final energy level is much lower than the reactants. Products: 2MgO (solid).

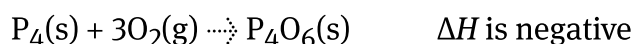
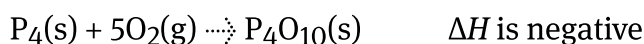
This graph demonstrates exothermic reactions: The negative enthalpy change (ΔH° equals to minus -285.8 kJ/mol means the reaction releases heat. Combustion makes the surroundings warmer. Magnesium burns in the presence of oxygen and results in a bright white flame, producing heat and light energy.

Combustion of non-metals

Conversely, many non-metals react with oxygen to form non-metallic oxides.

These oxides do not have fixed ratios, so these combustion reactions can produce several different product oxides for the same non-metal fuel.

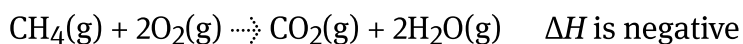
Take, for example, the reactions shown below between phosphorus and oxygen gas. Notice that both diphosphorus pentoxide and diphosphorus trioxide can be produced.



No matter the type of fuel, combustion reactions all involve at least one atom in the combustible material reacting with oxygen. This concept is covered in more detail in [subtopic R3.2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43486/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43486/), but it is still important to realise that oxygen is required for a combustion reaction to take place.

Combustion of organic compounds

In the complete combustion of methane gas, seen below, both the carbon and hydrogen atoms in the reactant molecules experience a gain in oxygen, and are fully oxidised.

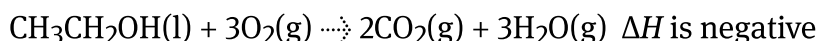


All organic compounds, especially those that contain only hydrogen and carbon (hydrocarbons), as well as alcohols, easily undergo complete combustion reactions that produce only carbon dioxide and water. Though these reactions always

require energy to break the bonds present between the atoms in the fuel and within molecules of oxygen gas, the products are always more energetically stable than the reactants. This means the overall complete combustion reaction is always exothermic!

It is important to realise that organic compounds are flammable no matter the state. It is possible to burn solid carbon (coal), so liquid and gas phase fuels are even more reactive. Always take care when working with fuels, as their vapours easily ignite.

Take for example the complete combustion of ethanol:



Rockets powered by combusting ethyl alcohol



Video 2. Combustion of organic alcohols are highly exothermic reactions.

Making connections

Ethanol can be blended at different ratios with gasoline to create what are known as flex fuels. Ethanol is added to gasoline in many countries to offset the need for fossil fuels. The addition of ethanol to gasoline allows for fewer carbon emissions, as well as a decrease in other types of air pollutants. You will have an opportunity to learn more about these types of fuels in section R1.3.4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/biofuels-id-45277/>), on biofuels.

Activity

- **IB learner profile attribute:**
 - Knowledgeable
 - Thinker
- **Approaches to learning:**
 - Thinking skills — Reflecting on the credibility of results
 - Research skills — Comparing, contrasting and validating information
- **Time required to complete activity:** 20 minutes
- **Activity type:** Pair activity

Instructions

Go on a fuel hunt!

Choose eight different substances — two reactive metals, two non-metals, two hydrocarbons and two alcohols — get creative!

For each substance:

1. Deduce the reaction equation for combustion when heated in oxygen.
2. Research the risks and benefits for the use of each of these substances as a fuel source and for the products formed in each reaction with respect to human health and the environment.
3. Determine which of these fuels would be your preferred choice.

Incomplete combustion of organic compounds

Learning outcomes

By the end of this section you should be able to:

- Deduce equations for the incomplete combustion of hydrocarbons and alcohols.
- Compare and contrast complete combustion with incomplete combustion of hydrocarbons and alcohols.

Many industries, places of work, and homes make use of the combustion of hydrocarbons. This can be to produce energy, for cooking, or in colder climates, as a source of heat. For safety reasons, buildings must be equipped with carbon monoxide detectors. How is it that the combustion of hydrocarbons can lead to the production of carbon monoxide, and why do we need to be concerned about such by-products?

When organic compounds undergo combustion in low-oxygen environments, they produce not only water and carbon dioxide gas, but other by-products. Depending on the amount of available oxygen, carbon monoxide gas and even carbon particles will be formed. This type of reaction is known as incomplete combustion. The lack of oxygen is often caused by the combustion reaction occurring too quickly, depleting the quantity of available oxygen around the burning fuel. Other factors that lead to incomplete combustion include, limited oxygen concentrations, low combustion temperatures, large fuel particles, and even the presence of water in solid fuels (like wood).

If the amount of oxygen gas is somewhat limited, carbon monoxide gas will form with water. Whereas if the amount of oxygen gas is extremely limited solid particles of carbon will form with water.

When observing a combustion reaction, we can often tell that the reaction is undergoing incomplete combustion due to it releasing smoke, and producing an orange or yellow flame. When a reaction has sufficient oxygen, the flame produced is much higher in temperature and appears blue in colour. As well, complete combustion reactions do not produce smoke.

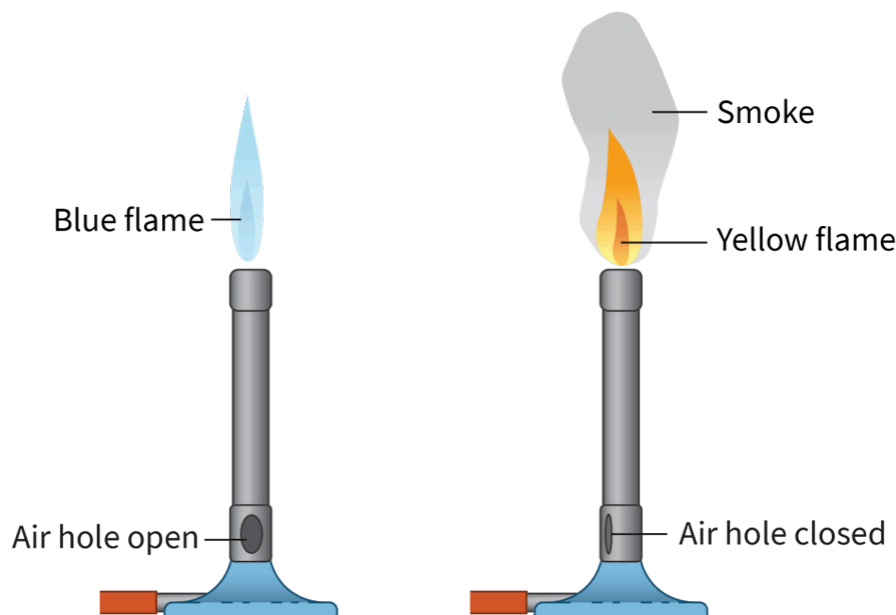


Figure 1. Complete vs. incomplete combustion of methane gas in a Bunsen burner.

 More information for figure 1

The image illustrates two Bunsen burners side by side. On the left, the Bunsen burner has an open air hole, producing a blue flame. This indicates complete combustion, which requires sufficient oxygen, resulting in a hotter, smokeless flame. On the right, the Bunsen burner's air hole is closed, creating a yellow flame and emitting smoke, demonstrating incomplete combustion due to limited oxygen supply. Key features such as 'Blue flame,' 'Yellow flame,' 'Smoke,' 'Air hole open,' and 'Air hole closed' are labeled next to their respective elements in the illustration.

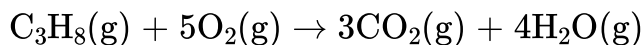
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Making connections

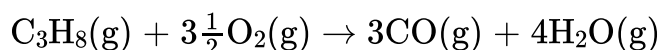
In sections S1.3.1 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/>) and S1.3.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-hydrogen-emission-spectrum-id-44042/>) you discovered that the wavelength of light produced in an emission spectrum is inversely related to the energy of the wave. Yellow and orange wavelengths of visible light (≈ 550 nm) have relatively long wavelengths and therefore low energies. Blue light has a shorter wavelength (≈ 430 nm) and therefore a higher energy.

Examples of complete and incomplete combustion per mole of propane combusted

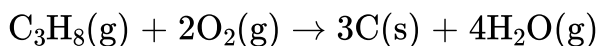
Sufficient oxygen supply:



Somewhat limited oxygen supply:



Extremely limited oxygen supply:



Nature of Science

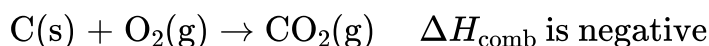
Aspect: Global impact of science

The carbon particles produced during incomplete combustion reactions often appear as black or grey smoke and stay as a suspension in the air. Known as sooty by-products, carbon particles that are aerosolised into the atmosphere lead to a reduction in air quality and can have major impacts on respiratory health.

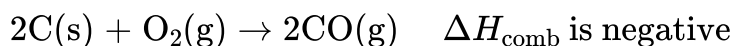
Carbon monoxide is an odourless, colourless, tasteless gas that is highly toxic and can lead to death if someone is exposed to even moderate concentrations for a prolonged period. Carbon monoxide molecules bind strongly to haemoglobin, in red blood cells, starving the body of oxygen.

The complete and incomplete combustion of carbon

Complete:



Incomplete:

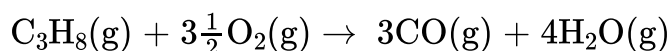


Both reactions involve the same reactants and are exothermic, but in the incomplete combustion reaction, the oxygen is highly limited and only carbon monoxide is formed as a product instead of carbon dioxide. The product produced contains a lower ratio of oxygen to carbon atoms.

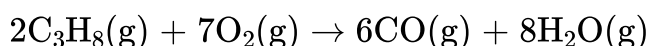
Incomplete combustion of hydrocarbons

All hydrocarbons undergo incomplete combustion reactions when reacted with limited amounts of oxygen gas. Take for example the incomplete combustion of propane, a common household and industrial fuel.

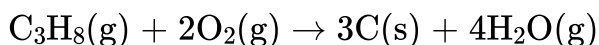
When the amount of oxygen is somewhat limited the reaction produces carbon monoxide gas and water vapour, as seen below:



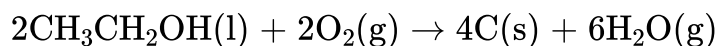
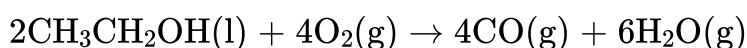
or balanced using whole-number coefficients:



When the amount of oxygen is extremely limited the reaction produces carbon particles and water vapour:



The same can be seen when looking at the incomplete combustion of alcohols. Take for example the incomplete combustion of ethanol:



Study skills

You have learned that the quantity of oxygen will determine the products of an incomplete combustion reaction. Some incomplete combustion reactions produce a combination of carbon products in varying ratios, depending on the amount of oxygen present. These specific reaction equations will be balanced for you since the ratio of products can differ from one reaction to the next.

Worked example 1

The incomplete combustion of propanol takes place under very low amounts of oxygen. List the products for this type of incomplete combustion reaction.

Solid carbon particulates and water vapour. These are the two products produced from the incomplete combustion of an alcohol (like propanol) under very low oxygen concentrations.

Somewhat low oxygen concentrations would lead to incomplete combustion, where the products would be carbon monoxide and water vapour.

Sufficient oxygen concentrations would lead to complete combustion, which has carbon dioxide and water vapour as the products.

Activity

- **IB learner profile attribute:**
 - Inquirer
 - Reflective
- **Approaches to learning:**
 - Communication skills — Using terminology, symbols and communication conventions consistently and correctly.
 - Thinking skills — Using terminology, symbols and communication conventions consistently and correctly,
 - Research skills — Comparing, contrasting and validating information
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Pair activity

Instructions

In this activity you are asked to work with a partner to perform research on carbon monoxide detectors. You will need to use two or more reliable sources to support your answers to the following questions. The answers to these questions will help you to connect the topic of incomplete combustion to real life situations.

Recall that a reliable source is an article, journal, report, website etc. where you have confidence in the validity of the material. Examples include textbooks, reference books and databases, as well as government research documents and peer-reviewed reports.

1. List a variety of locations where it is necessary to install a carbon monoxide detector.
2. Describe the chemical(s) that these devices detect.
3. Discuss a particular type of carbon monoxide detector and how it functions. Identify the sensitivity of the chosen detector.
4. Explain why these types of detectors are essential in certain situations (types of heating) and not others. Support your answer by linking it to the incomplete combustion reaction for propane (a common heating fuel).

Fossil fuels

Learning outcomes

By the end of this section you should be able to:

- Evaluate the amount of carbon dioxide added to the atmosphere when different fuels burn.
- Compare the relative amounts of energy released per unit mass of different fossil fuels and their tendency to undergo incomplete combustion.
- Identify the relationship that exists between carbon dioxide levels and the greenhouse effect.

Not all fuels are pure substances like the example of the combustion of propane given in the previous section. Some fuels are mixtures of many compounds. Fossil fuels can contain a variety of compounds that formed through the decomposition of prehistoric organisms over time, deep within the Earth's crust. The use of fossil fuels for heat and energy is widespread in homes and businesses. How might the combustion of fossil fuels comprising a range of compounds differ compared to the burning of those that contain only one fuel source, such as propane?



Figure 1. Fossil fuels provide nearly 85% of the world's total energy.

Credit: Schropfschop, Getty Images

Fossil fuels

Fossil fuels are subdivided into a variety of categories based on the type of prehistoric organism they formed from and their physical and chemical properties. Fossil fuels are largely divided into three main categories; coal, crude oil, and natural gas (**Table 1**). The components of fossil fuel mixtures can be separated by their boiling point in a process called fractional distillation.

Table 1. Features of fossil fuels.

Type of fossil fuel	State at room temperature	Composition
Coal	Solid	Mainly carbon, traces of hydrogen, oxygen, nitrogen, sulfur and metals.
Crude oil	Liquid	Mixture of medium to long chain hydrocarbons, traces of sulfur, nitrogen and oxygen and metals.
Natural gas	Gas	Mostly methane, traces of other small hydrocarbons, water and carbon dioxide.

Advantages and disadvantages of fossil fuels

Fossil fuels are highly exothermic when combusted and are relatively easy to transport, making them convenient to use in many industries. Also, fossil fuels are abundant in many countries and are relatively inexpensive to obtain compared to the preliminary set-up cost of other types of energy resources, such as nuclear power. Fossil fuels contribute to the overall welfare of people and the economy by providing jobs, income and taxes for many countries and regions. They also support other industries, such as agriculture, manufacturing and power generation.

Conversely, fossil fuels have many disadvantages. The mining, transportation, and processing of fossil fuels are damaging to both local and global environments, further enhanced by accidents such as oil spills, pipeline leaks and coal mine explosions. The burning of fossil fuels produces substances that are harmful to human health and the environment, contributing to global warming, respiratory problems and the acidification of lakes and soil. Since fossil fuels cannot be synthesised, our planet has exhaustible stores of these non-renewable resources, making extraction and processing more challenging.

Making connections

Due to the combination of elements and compounds found in many fuels, the combustion of fuels like coal, biofuel, gasoline, and wood can produce a variety of carbon oxides, sulfur oxides (SO_x) and oxides of nitrogen (NO_x). The contribution of these products to the acidity of rain and other forms of precipitation is further described in [section S3.1.5](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-and-nonmetallic-continuum-id-44695/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-and-nonmetallic-continuum-id-44695/>).

Greenhouse effect

Of major importance is the environmental impact of the combustion of fossil fuels. The products of combustion – carbon oxides and other oxides – are released into atmosphere. With the widespread use of fossil fuels, what effect does this have?

Examine **Figure 2** below. It shows the concentration of CO₂ in the atmosphere over a prolonged period of time. What do you notice about the data?

There is a regular cyclic pattern in the data dating back over thousands of years. The CO₂ levels have naturally risen and fallen throughout history. Look at the far right of the chart where the data shows a rapid rise in CO₂ levels. This rapid rise has been determined to be mostly caused by human's use of fossil fuels, producing the highest levels of CO₂ in the atmosphere.

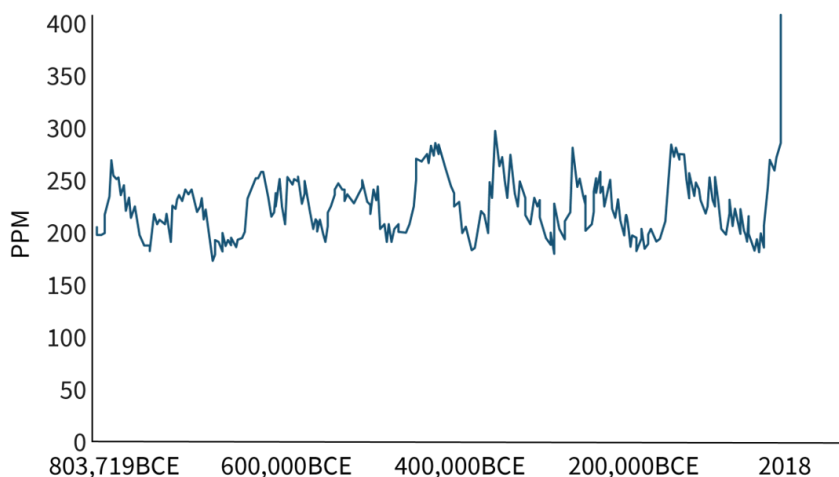


Figure 2. Atmospheric carbon dioxide levels through history.

 More information for figure 2

The image is a line graph depicting atmospheric carbon dioxide levels over time, extending back to 803,719 BCE and continuing to the year 2018. The X-axis represents time in years, marked at intervals of 803,719 BCE, 600,000 BCE, 400,000 BCE, 200,000 BCE, and 2018. The Y-axis represents CO₂ levels in parts per million (PPM), ranging from 0 to 400. Throughout history, the graph shows a regular cyclic pattern, with levels consistently rising and falling. A notable feature is the sharp rise in CO₂ levels at the far right of the chart, indicating a significant increase in recent times, attributed mainly to human activity and fossil fuel consumption.

[Generated by AI]

Now examine **Figure 3**, which contains the average global temperature over the last two centuries. What similarities do you see with **Figure 2** above?

There is a cyclic pattern to the global temperature, just as with the CO₂ levels. There is also a rapid increase in global temperature in more recent years. It appears that the two variables – CO₂ levels and global temperature – are related. Why do you think this is so?

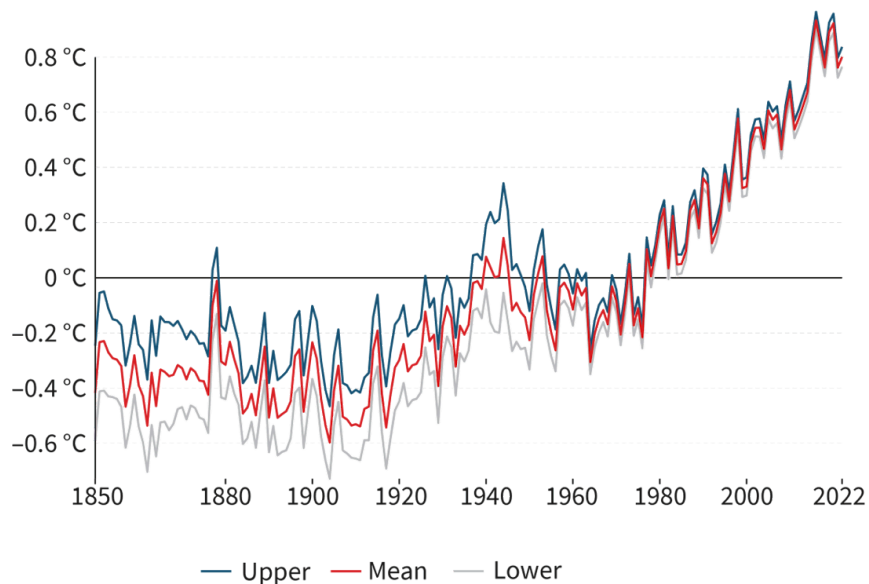


Figure 3. Global temperature through history.

 [More information for figure 3](#)

The graph depicts global temperature changes from 1850 to 2022. The Y-axis indicates temperature deviations in degrees Celsius ranging from -0.6°C to $+0.8^{\circ}\text{C}$. The X-axis spans the years 1850 to 2022. Three lines are shown: Upper, Mean, and Lower. From 1850 to around 1930, temperatures showed fluctuations with several peaks and troughs but remained below 0°C . From 1930 onwards, temperatures began rising, with notable peaks around 1940, 1980, and reaching a rapid increase after 2000. By 2022, temperatures reached around $+0.8^{\circ}\text{C}$, indicating a clear upward trend in global temperatures over the years.

[Generated by AI]

The greenhouse effect is a phenomenon that occurs when certain gases, such as carbon dioxide, methane or nitrogen oxides, absorb infrared (IR) radiation that has been emitted from the Earth's surface after it has been heated by the Sun's rays. The greenhouse gases then emit the IR radiation in all directions so the net effect is that a slightly higher proportion of the solar energy is trapped within the atmosphere, leading to higher average global temperatures. Since the combustion of fossil fuels releases carbon dioxide gas, their continued use contributes to the greenhouse effect.

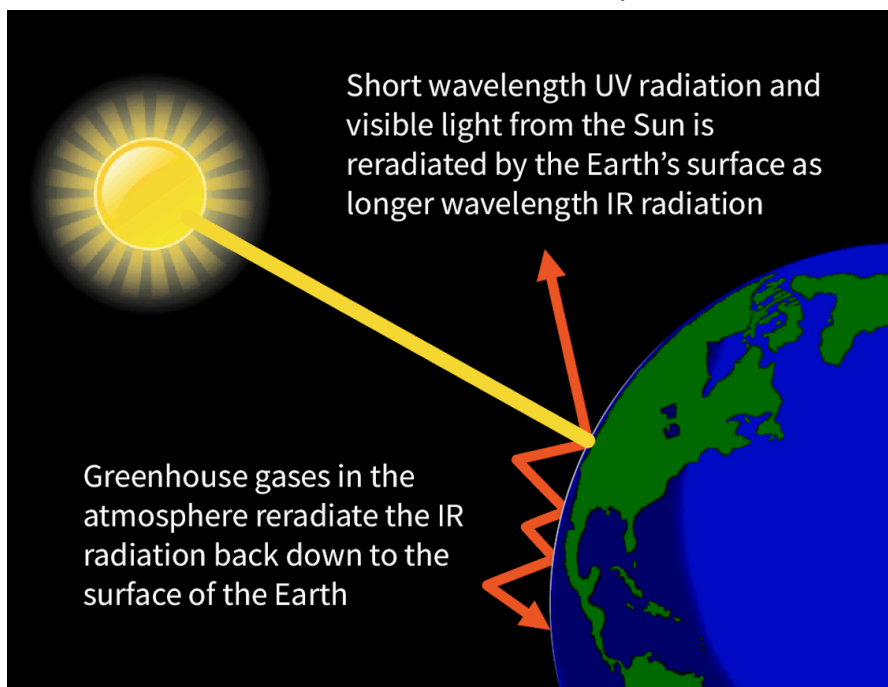


Figure 4. The relationship between atmospheric carbon dioxide levels and the greenhouse effect.

Source: adapted from 'Climate Change Schematic' [↗](#)

(https://commons.wikimedia.org/wiki/File:Climate_Change_Schematic.svg)

by Efbrazil, licensed under CC BY-SA 4.0 [↗](#)

(<https://creativecommons.org/licenses/by-sa/4.0/deed.en>)

[🔍](#) More information for figure 4

The image consists of a diagram illustrating the greenhouse effect. It shows a representation of the Earth with solar radiation depicted as a thick yellow line coming from the sun towards the Earth. Some of this radiation is re-radiated back into space as illustrated by a smaller red arrow noting 'Short wavelength UV radiation and visible light from the Sun is re-radiated by the Earth's surface as longer wavelength IR radiation'. Some radiation is also re-emitted towards Earth, as indicated by a zig-zagging red arrow noting 'Greenhouse gases in the atmosphere re-radiate the IR radiation back down to the surface of the Earth'.

International Mindedness

Fossil fuels like coal and crude oil naturally contain sulfur and nitrogen atoms due to the residues present in the proteins of the pre-historic biological materials that formed these deposits. When fossil fuels are combusted, they produce not only carbon oxides, and water vapour, but also sulfur oxides (SO_x) and oxides of nitrogen (NO_x). Both nitrogen dioxide and sulfur dioxide contribute to acid deposition as is seen in [section S3.1.5](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-and-nonmetallic-continuum-id-44695/>). Due to prevailing winds, the air pollution produced in one country or region flows into other countries and territories. The products of the combustion of fossil fuels is a global issue and one we must address as a global community.

Amount of carbon dioxide produced

All fossil fuels release carbon dioxide when they undergo combustion, but the amount varies depending on the fuel type.

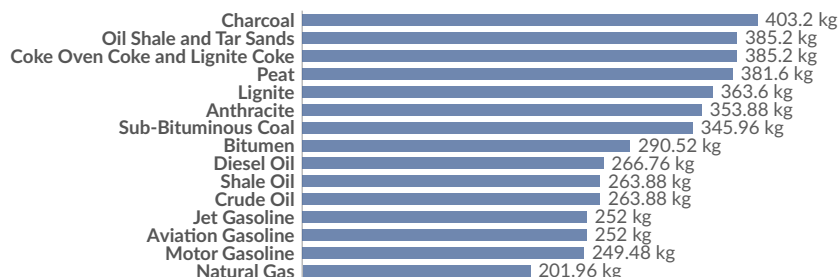
Carbon dioxide emissions factors

Emissions factors quantify the average CO₂ output per unit of energy. They are measured in kilograms of CO₂ per megawatt-hour (MWh) of energy from various fossil fuel sources.

Our World
in Data

Table Chart

Edit



Data source: IPCC - Emission Factor Database

(2025) - [Learn more about this data](#)

OurWorldinData.org/fossil-fuels | CC BY



Explore the data →

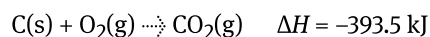
Figure 5. The amount of carbon dioxide released per unit energy of fuel.

More information for figure 5

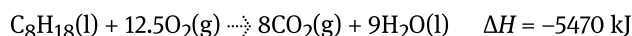
The interactive helps students assess the environmental impact of burning various fuels by assessing CO₂ emissions, comparing energy outputs and combustion efficiencies, and examining how these factors contribute to the greenhouse effect.

It features a horizontal bar chart that displays the carbon dioxide emission factors, measured in kilograms per megawatt-hour, for various fossil fuel sources. The y-axis represents values in kilograms. The x-axis represents various fossil fuel sources. The values, in decreasing order, are provided in a separate tab titled table.

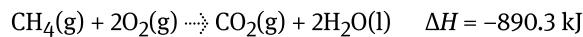
Coal, consisting mainly of carbon, produces a high quantity of carbon dioxide during its combustion, as shown in the reaction equation. Since there is only one product formed in this reaction, the quantity of CO₂ is high compared to the amount of energy released.



Crude oil, a mixture of longer chain hydrocarbons, produces carbon dioxide and water during combustion, as shown in the reaction equation below for octane, one of the components of crude oil. Since there are two products formed, the amount of CO₂ produced per unit of energy released is lower than coal.



Natural gas, consisting mostly of methane, also produces carbon dioxide and water, as shown in the reaction equation below. Just as with crude oil above, the formation of two products lowers the ratio of CO₂ formed per unit of energy. Natural gas has a lower carbon to hydrogen ratio compared to longer hydrocarbon chains; 1 C: 4 H in methane, compared to 1 C: 2.25 H in octane, resulting in less CO₂ released per unit energy compared to crude oil.



These three fuel sources can be compared with the amount, in mol (or by volume), of CO_2 produced per unit of heat (**Table 2**).

Table 2. CO_2 per unit of energy from fuel sources.

Type of fuel	Reaction equation	Mol CO_2 per MJ of energy
Coal	$\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$	$\frac{1 \text{ mol CO}_2}{-393.5 \text{ kJ}} \times 1000 = 2.54 \text{ mol CO}_2 \text{ MJ}^{-1}$
Crude oil	$\text{C}_8\text{H}_{18}(\text{l}) + 12.5\text{O}_2(\text{g}) \rightarrow 8\text{CO}_2(\text{g}) + 9\text{H}_2\text{O}(\text{l})$	$\frac{8 \text{ mol CO}_2}{-5074 \text{ kJ}} \times 1000 = 1.58 \text{ mol CO}_2 \text{ MJ}^{-1}$
Natural gas	$\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$	$\frac{1 \text{ mol CO}_2}{-890.3 \text{ kJ}} \times 1000 = 1.12 \text{ mol CO}_2 \text{ MJ}^{-1}$

Therefore, coal produces the greatest amount of CO_2 per unit energy, followed by crude oil and then natural gas.

If the reaction occurs at standard temperature and pressure, we can use the mole ratios and molar volume of gas produced in the reaction.

Table 3. Molar volume of CO_2 produced from fuel sources.

Type of fuel	Amount CO_2 produced	Mass of fuel	Volume per fuel
Coal	$1 \text{ mol} \times 22.7 \text{ dm}^3 \text{ mol}^{-1} = 22.7 \text{ dm}^3$	$1 \text{ mol of C} = 12.01 \text{ g}$	$\frac{22.7}{12.01} = 1.8$
Crude oil	$8 \text{ mol} \times 22.7 \text{ dm}^3 \text{ mol}^{-1} = 181.6 \text{ dm}^3$	$1 \text{ mol C}_8\text{H}_{18} = 114.26 \text{ g mol}^{-1}$	$\frac{181.6}{114.26} = 1.6$

Type of fuel	Amount CO ₂ produced	Mass of fuel	Volume per fuel
Natural gas	$1 \text{ mol} \times 22.7 \text{ dm}^3 \text{ mol}^{-1} = 22.7 \text{ dm}^3$	$1 \text{ mol CH}_4 = 16.05 \text{ g}$	$\frac{22.7}{16.05} = 1.4$

Therefore, coal also produces the largest volume of carbon dioxide per unit mass of fuel, followed by crude oil and natural gas produces the least.

Theory of Knowledge

You can see real time air quality values for many cities in the world at <https://waqi.info/>  (<https://waqi.info/>). The laws promoting and enforcing clean air vary from country to country, but the air is a shared resource. Who has the responsibility for the quality of the air we breathe?

Energy released per unit mass

All fossil fuels release energy when they undergo combustion, but the amount varies depending on the fuel type. The amount of energy can be compared per mole in thermochemical equations for a specific substance undergoing combustion. Since the fossil fuel types are mixtures and vary in composition, the amount of energy released per unit mass of fuel is often used for comparison, and these values can vary as well. As you can see in **Table 3**, coal produces the least amount of energy per unit mass, followed by crude oil and natural gas.

Table 3. Energy per unit mass from fuel sources.

Type of fuel	Energy per unit mass (MJ kg ⁻¹)
Coal	29.3
Crude oil	42.3

Type of fuel	Energy per unit mass (MJ kg ⁻¹)
Natural gas	50.0

Tendency for incomplete combustion

As you have seen, in [section R1.3.2](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/incomplete-combustion-of-organic-compounds-id-45275/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/incomplete-combustion-of-organic-compounds-id-45275/>) incomplete combustion reactions lead to the formation of carbon monoxide gas and not carbon dioxide gas. All forms of fossil fuels are at risk for incomplete combustion if the air to fuel ratio is not optimised. Consequences of incomplete combustion are:

- less energy released per mole of fuel in comparison to complete combustion reactions
- harmful by-products, including carbon monoxide gas and carbon (soot) particulates
- unreacted fuel can pose a fire/explosion hazard.

Coal has the greatest tendency to undergo incomplete combustion due to the presence of impurities such as nitrogen, sulfur and other metals that also react with oxygen. Since these impurities consume oxygen, there may not be enough oxygen left to react with the carbon in the coal, leading to incomplete combustion.

Crude oil and natural gas mostly consist of hydrocarbons – crude oil with longer chains and natural gas mostly methane. Longer hydrocarbon chains require more oxygen and are therefore more likely to undergo incomplete combustion than shorter hydrocarbon chains. Therefore, natural gas is the least likely fossil fuel to undergo incomplete combustion.

Creativity, activity, service

Strand: Service

Learning outcome: Demonstrate engagement with issues of global significance

The quality of the air that we breathe can have a significant effect on our health. Particulate matter [🔗](https://www.epa.gov/pm-pollution/particulate-matter-pm-basics) (<https://www.epa.gov/pm-pollution/particulate-matter-pm-basics>) (PM) is very small airborne particles such as soot, dust or smoke. Sources of PM include the burning of gasoline or diesel fuel in internal combustion engines and the combustion of wood. PM can be classified as PM₁₀ or PM_{2.5} depending on the size of the particles. Short-term health effects of PM_{2.5} include irritation to the eyes, nose or lungs. Long-term effects include premature death in those people who suffer from pre-existing health conditions.

A possible CAS experience could be to determine the PM_{2.5} levels in your [region](https://www.stateofglobalair.org/air/pm) [🔗](https://www.stateofglobalair.org/air/pm) (<https://www.stateofglobalair.org/air/pm>) and raise awareness of the possible health effects. You could also research ways in which the levels of PM_{2.5} can be reduced and share these with your school or local community.

In the activity below you will use an interactive to help you to make connections between global atmospheric temperatures and atmospheric concentrations of carbon dioxide.

Activity

- **IB learner profile attribute:** Thinker
- **Approaches to learning:** Thinking skills
 - Experimenting with new strategies for learning
 - Combining different ideas in order to create new understandings
- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual or pairs activity

Instructions

In this section you have studied the relationship between carbon dioxide emissions and the greenhouse effect. Using the following interactive simulation, you will look at modelling the relationship between atmospheric carbon dioxide concentrations and other dependent variables like atmospheric temperature, oceanic carbon dioxide concentration, and atmospheric water vapour concentrations.

<https://has.concord.org/global-climate.html>  (<https://has.concord.org/global-climate.html>)

1. Create a data table containing space for three different concentrations of atmospheric carbon dioxide (50%, 100% and 150% of 2010 levels).
2. Adjust the level of Human Emissions (% of 2010 values) for carbon dioxide by clicking and moving the button to the left or right. This will allow you to set the three different concentrations of carbon dioxide you chose.
3. Run the simulation for at least a 25-year period.
4. Record the approximate temperature change for the atmosphere, for each concentration of carbon dioxide that you chose.
5. Describe the trend(s) you observe between atmospheric carbon dioxide levels and atmospheric temperature. Support your answer with information from this unit.
6. To what extent does the combustion of fossil fuels relate to the increase in atmospheric temperatures?
7. Examine the relationship that exists between the type of fossil fuel combusted (coal, crude oil, natural gas) and the quantity of carbon dioxide released into the atmosphere.
8. Extension question: Interpret the relationship that is shown in the simulation between carbon dioxide concentrations and oceanic carbon dioxide concentration.

Biofuels

Learning outcomes

By the end of this section you should be able to:

- Summarise the reactants and products of photosynthesis.
- Contrast renewable and non-renewable energy sources.
- Recognise the advantages and disadvantages of biofuels.

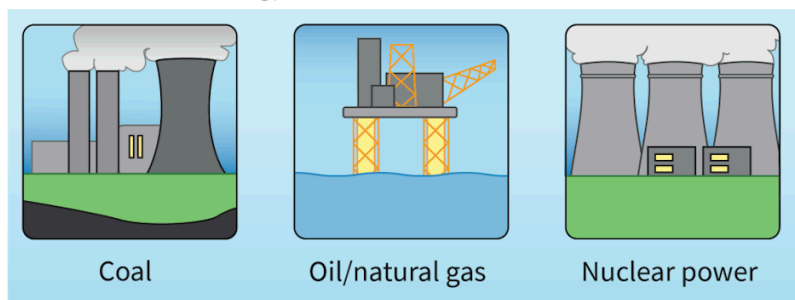
The Sun's energy is the driving force for nearly all energy cycles on Earth. For billions of years, life on Earth has evolved to best harness this practically limitless source of energy. Without the ability of plants, fungi, and other organisms to convert solar radiation to chemical energy, life on Earth would not have developed as we know it. The process by which living organisms convert solar energy to chemical energy is called photosynthesis. But how does this relate to chemistry? And how can we harness this limitless energy resource to meet our current and future energy requirements?

Renewable vs. non-renewable energy sources

All energy sources are classified as being renewable or non-renewable. A non-renewable energy source is consumed in a chemical or nuclear process and cannot be renewed at a rate equal to or faster than it is consumed. Energy resources like enriched uranium, which is used in fission nuclear reactors, and fossil fuels are non-renewable.

On the other hand, energy resources like solar energy, hydroelectricity, and wind power are classified as renewable. Other renewable resources like biofuels, and hydrogen fuel cells are more closely connected to chemical processes, and you will learn more about them in this section and section R1.3.5 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fuel-cells-id-44783/>) respectively.

Non-renewable energy



Renewable energy

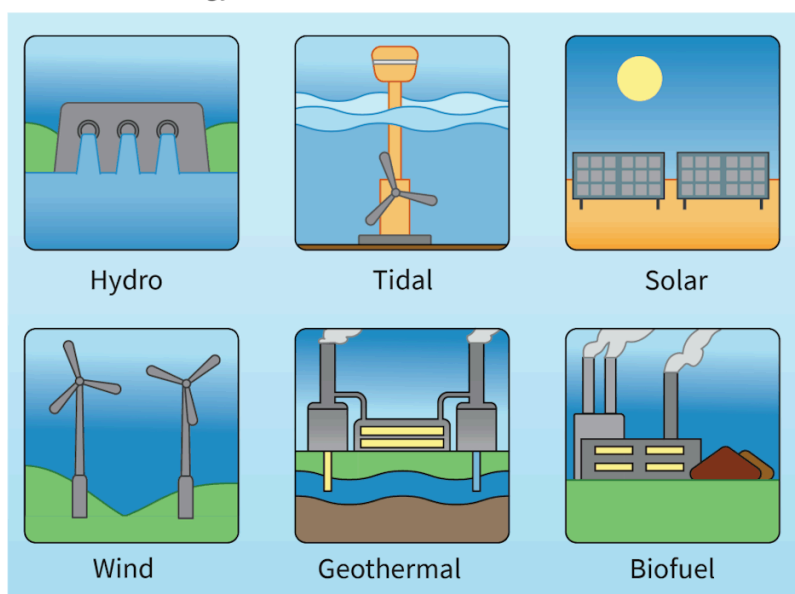


Figure 1. Renewable vs. non-renewable energy resources.

Renewable energy resources are often confused with green energy – energy sources that have a reduced environmental impact. In fact, both renewable and non-renewable energy resources have advantages and disadvantages much like the combustion of fossil fuels studied in [section R1.3.3](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fossil-fuels-id-45276/>). Use **Interactive 1** to sort the energy resources into renewable energy and non-renewable energy.

Renewable	Non-renewable











✓ Check

📄 Rights of use

Interactive 1. Renewable vs. Non-Renewable Energy Resources.

Photosynthesis and the production of biofuels

Biofuels are, as the name suggests, fuels that are derived from biological sources. These fuels, like those studied in [section R1.3.3](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/fossil-fuels-id-45276/>), are organic molecules that can be combusted, and the energy released is harnessed for use in many different applications. What exactly is a biofuel, and how are these fuels formed?

To have a better understanding of biofuels we need to start at the beginning, which in this case is the Sun! The Sun emits electromagnetic radiation at various wavelengths, as seen in [section S1.3.1](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/line-spectra-id-43112/>), which corresponds to different levels of energy. Plants and other organisms harness this energy by absorbing certain wavelengths of light within their chlorophyll-containing pigments. The wavelengths of light that are absorbed by chlorophyll generate electrons to power photosynthesis.

The chlorophyll-containing pigments then produce glucose, which can be used by the plant for immediate energy or further transformed and stored. These photo-chemical reactions also require carbon dioxide gas and water vapour as the

primary reactants and produce oxygen gas as a by-product. This highly complex biological pathway, which consists of many different steps and mechanisms, is often reduced to a simple equation.

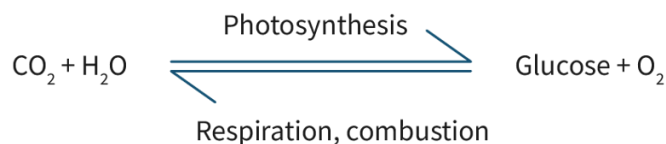


Figure 2. Photosynthesis and combustion.

 More information for figure 2

The diagram illustrates the processes of photosynthesis and combustion through two arrows. On the left side of the diagram, there are the chemical compounds "CO₂ + H₂O". An upward arrow is labeled "Photosynthesis," pointing from the left to the right, leading to "Glucose + O₂" on the right side. Below this, a downward arrow labeled "Respiration, combustion" points back from "Glucose + O₂" to "CO₂ + H₂O," showing the reverse process. This visual representation highlights the cyclic nature of these biological and chemical processes: carbon dioxide and water being converted into glucose and oxygen through photosynthesis, and then glucose and oxygen converting back into carbon dioxide and water through respiration or combustion.

[Generated by AI]

Some glucose molecules are used immediately by the plant for its energy needs, while others are converted to cellulose or polysaccharides like starch or into lipids to help with structure and energy storage. These molecules are not only used by plants for the storage of energy, but can be used by humans to generate energy as well. We do this by converting these molecules into combustible materials called biofuels. As the name implies, biofuels are fuels that are derived from biotic or living sources.

From plant material to fuels

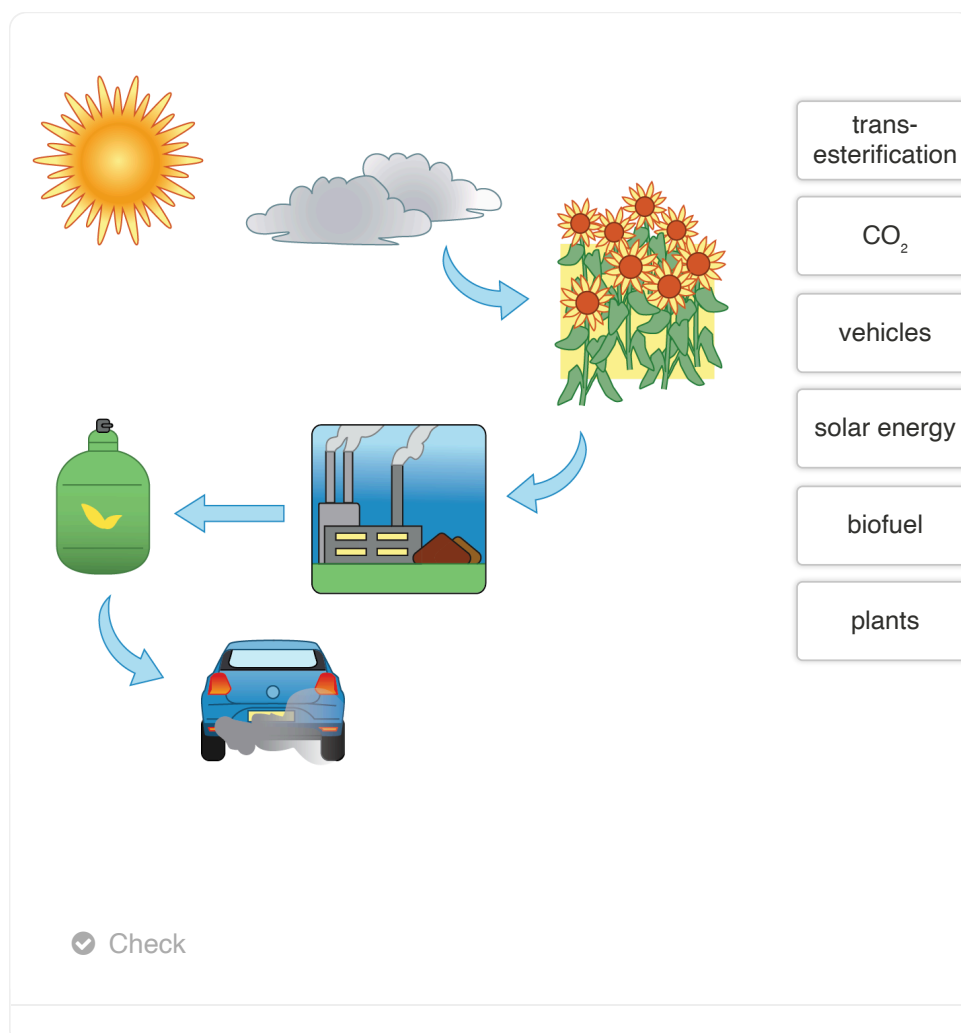
Biofuels are not a recent discovery and have been used as a form of energy by humans for thousands of years. If you think about it, humans have been burning plant-based oils and materials like wood since prehistoric times.

Modern day biofuels are derived from plant material that is transformed via physical and chemical processes to produce usable fuels. We then combust these biofuels in the presence of oxygen gas, producing carbon dioxide, water, and energy. The biofuels we use today, like ethanol and biodiesel, are more processed

and are derived from many different types of organisms; from sugar-rich crops like corn and sugar cane to cellulose, food, and animal waste, and more recent sources like algae and cyanobacteria.

No matter the primary source, the biofuels are produced via the breakdown or decomposition of larger organic molecules into combustible short-chain hydrocarbons like alkanes, alcohols, glycerol, and ethyl esters.

Use **Interactive 2** to complete the pathway representing the conversion of sunlight to biofuels.



Interactive 2. A pathway representing the conversion of sunlight to biofuels.

More information for interactive 2

An interactive illustration displays the conversion of sunlight to biofuels using arrows to show each step in the cycle. Users need to drag and drop the correct terms into the appropriate drop zones that correspond to each stage of the process.

The sequence in the illustration includes the following elements:

- The Sun
- Clouds
- Plants with flowers

- An industry/processing plant
- A green fuel can with two leaves
- A car with smoke coming out of the exhaust

The drag-and-drop options provided are:

Transesterification, CO₂, Vehicles, Solar energy, Biofuel, Plants

This interactivity helps users understand the transformation of solar energy into biofuels by engaging them in a visual and hands-on activity.

It introduces solar energy as the primary source and illustrates how plants absorb carbon dioxide and sunlight for growth through photosynthesis. Users learn about transesterification as the key chemical process that converts plant oils into usable biofuels. The final step connects biofuels to their real-world application in vehicles, highlighting the practical use of renewable energy.

Read below for the solution:

Sun- Solar energy

Clouds- CO₂

Plant with flowers- Plants

Industry- Transesterification

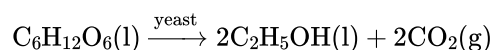
Green fuel can with two leaves - Biofuel

Car with smoke coming out of the exhaust- Vehicles

International Mindedness

Most countries, in their efforts to improve fuel security, have in place research and development programs for the production and distribution of biofuels.

In the latter half of the twentieth century, Brazil became one of the pioneers in the production of bioethanol. Bioethanol is a useful biofuel, which can be added to gasoline to produce flex fuels. Bioethanol is produced via the fermentation of plant material (feedstock) that is high in starch and sugars. This process takes place in anaerobic conditions in the presence of yeast, which act as a biological catalyst.



Many companies, around the globe, are also using food waste to produce biodiesel. This is achieved through the conversion of plant-based lipids (triglycerides) to glycerol and methyl ester through a reaction known as transesterification. The reaction needs to be carried out in the presence of a strong base and methanol or ethanol, and in the end old vegetable oil and grease can be converted into a usable fuel.

Advantages and disadvantages of biofuels

Biofuels have many advantages. The plants used to make biofuels must absorb carbon dioxide when performing photosynthesis, which offsets the carbon emissions of the biofuel and it undergoes combustion. As well, since some biofuels are short-chain hydrocarbons, the combustion reactions they undergo are

cleaner than traditional fossil fuels. As you may recall from earlier in this section, short-chain hydrocarbons undergo complete combustion more readily than long chain hydrocarbons.

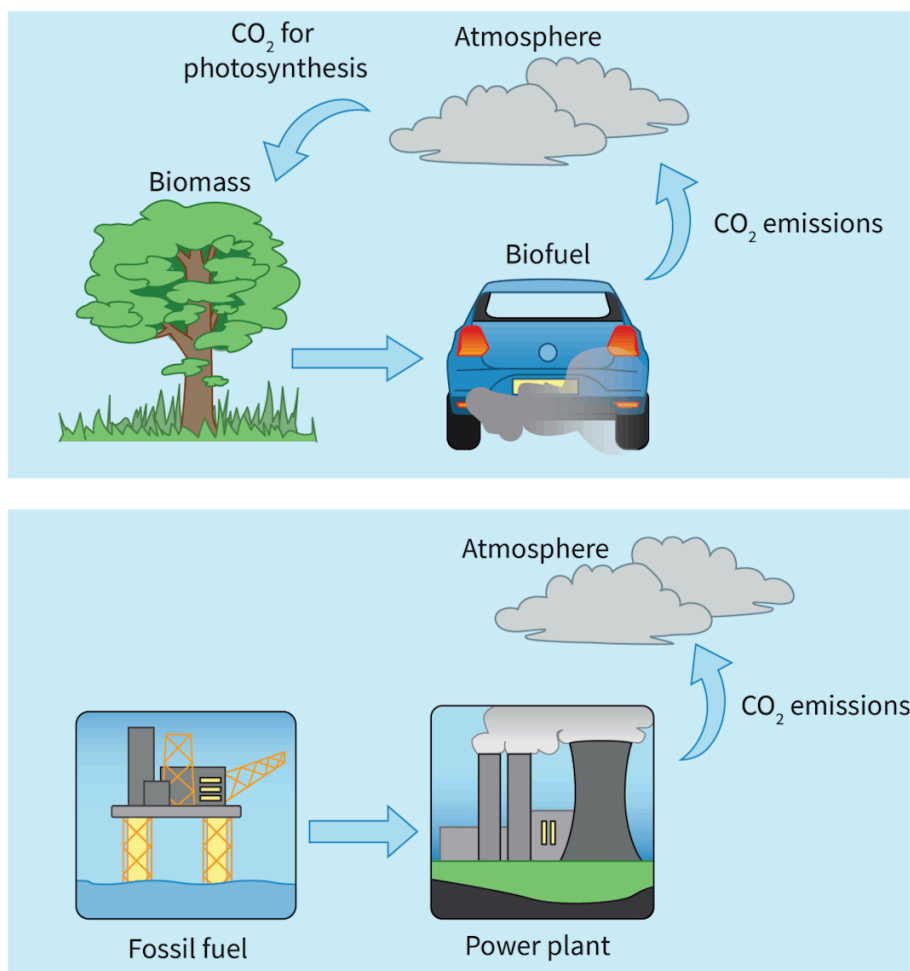


Figure 3. Partial offsetting with biofuels.

 More information for figure 3

The image is divided into two sections to illustrate the processes of biofuels and fossil fuels.

The top section shows the biofuel cycle. On the left, there is a graphic of a tree labeled "Biomass." An arrow points from the tree to a car labeled "Biofuel." This indicates the conversion of biomass into biofuel. The car emits "CO₂ emissions" depicted with an arrow pointing to a cloud labeled "Atmosphere." Another arrow circles back from the atmosphere to the tree with the text "CO₂ for photosynthesis," illustrating how plants absorb carbon dioxide, thus partially offsetting emissions.

The bottom section portrays the fossil fuel process. On the left, there is an image of an oil platform labeled "Fossil fuel." An arrow leads to a power plant on the right labeled "Power plant." Above the power plant, smoke is shown with an arrow pointing to another cloud labeled "Atmosphere," indicating "CO₂ emissions" from burning fossil fuels, which contribute to atmospheric pollution.

The diagram contrasts the carbon cycle in biofuels versus fossil fuels, emphasizing biofuels' less harmful environmental impact.

[Generated by AI]

In addition to this, the refining and synthesis process is less harmful to the environment, using fewer resources than fossil fuel refining. There is no need to mine these fuels or use fractional distillation in their production. Biofuels are relatively inexpensive to produce compared to other types of renewable energy resources, and the raw materials to make them are readily available. Lastly, by acting as a fossil fuel substitute, biofuels allow us to preserve our fossil fuel stores for future use in other applications.

On the other hand, biofuel production uses large quantities of plant material as its feedstock, which requires land that is appropriate for agriculture. Though the combustion of biofuels is somewhat less harmful than the combustion of traditional fossil fuels, the products produced in the combustion of biofuels still emit greenhouse gases into the atmosphere (**Table 1.**).

Table 1. Advantages and disadvantages of biofuels.

	Biofuels
Advantages	• Release less CO₂ compared to fossil fuels • Inexpensive • Can be used directly after synthesis — no mining or processing at a refinery as with fossil fuels
Disadvantages	• Uses land suitable for agriculture • Combustion still releases CO₂ into the atmosphere • Growth of crops requires the use of fossil fuels

Activity

- **IB learner profile attribute:**
 - Communicator
 - Thinker
- **Approaches to learning:**
 - Thinking skills — Evaluating and defending ethical positions
 - Communication skills — Clearly communicating complex ideas in response to open-ended questions, Using digital media for communicating information
- **Time required to complete activity:** 30–40 minutes

- **Activity type:** Individual activity

Instructions

In this section you explored the nature of renewable and non-renewable energy resources, and the origin of these sources of energy. In this activity you are encouraged to explore the production of biofuels in your region or country and to evaluate the ethical implications of investing in these types of fuel sources. Write a reflection on the process as well as the information you researched.

Many countries are turning to biofuels to meet their energy needs. Since the early 1970s Brazil has been investing heavily in the development of their biofuel industry.

1. Research what types of biofuels are produced in your region or country. What are the sources of these biofuels and how are they currently being distributed and used?
2. Evaluate what some of the benefits and drawbacks might be for a country investing in this type of energy resource. These benefits/drawbacks could be environmental, social, political, etc.

Fuel cells

Learning outcomes

By the end of this section you should be able to:

- Compare and contrast hydrogen and methanol fuel cells.
- Deduce half-equations for the electrode reactions in a fuel cell.

Around the globe, many countries are turning to renewable and/or green energy sources to try to solve the world's energy needs, while still addressing our ever-increasing concerns about the greenhouse effect and global climate change. Some countries have the natural resources and infrastructure necessary to make this transition, but other countries do not. In some cases, countries still rely heavily on the combustion of fossil fuels and biofuels to generate electricity. But what other ways are there to derive energy from molecules? Is there any way to avoid the combustion of fuels to meet our energy requirements?

Fuel cells vs. conventional techniques

Electricity can be produced from the combustion of fossil fuels such as coal and natural gas. In these power plants, the heat released from the combustion reaction undergoes many transfers and transformations to become electrical energy. In a fuel cell, electrical energy is derived directly from a chemical reaction without the need for multiple transformations of energy. As such, fuel cells are highly energy efficient compared to their classic counterparts.

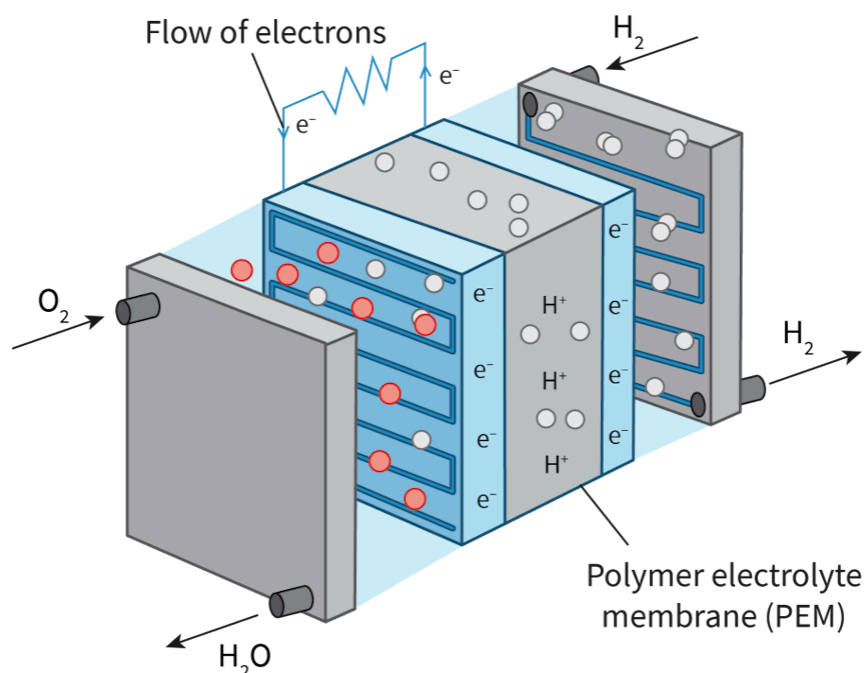


Figure 1. A simple fuel cell.

 More information for figure 1

The diagram shows a simple fuel cell that includes several key components and labels. On the left, oxygen (O_2) is depicted entering the cell. On the right, hydrogen (H_2) is shown entering from two directions. Water (H_2O) is indicated as a product flowing out.

The central part of the diagram consists of a polymer electrolyte membrane (PEM), which separates the cell into two regions. These regions are connected by a pathway labeled as the flow of electrons (e^-), representing the movement of electrons from one side of the membrane to the other.

The diagram illustrates ions and electrons participating in the reactions. H_2 splits into H^+ ions and electrons (e^-) at one electrode. These electrons travel through an external circuit path, indicated by a zigzag line labeled "Flow of electrons," to provide electrical power.

Hydrogen ions (H^+) move through the PEM to meet oxygen at the other electrode, forming water (H_2O) as a byproduct. The entire process illustrates the conversion of chemical energy to electrical energy in a fuel cell.

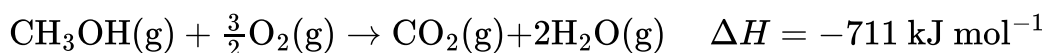
[Generated by AI]

A fuel cell consists of a spontaneous exothermic chemical reaction that is divided in two regions. These two half-reactions occur separately in two different locations, separated by a membrane, but connected by a wire. The wire allows for the flow of electrons from one half of the reaction to the other.

Electrons are formed at the side of the chemical reaction where oxidation is occurring. The electrons flow into the wire and then towards the other side of the fuel cell where they are used. The movement of electrons through the wire creates an electrical current that can be used to directly charge a load or operate a piece of machinery.

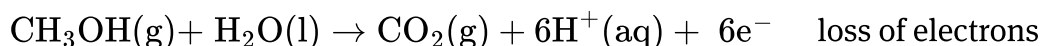
Methanol fuel cells

Let us take as an example a combustion reaction that you studied in [section R1.3.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/combustion-reactions-id-43476/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/combustion-reactions-id-43476/>). Here we are combusting methanol in the presence of oxygen gas to produce carbon dioxide and water vapour.

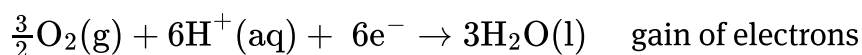


A methanol fuel cell works with the same chemical reaction but separates the process into two half-reactions.

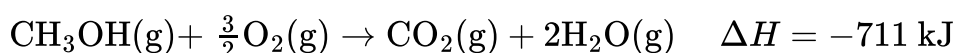
Half reaction:



Half reaction:

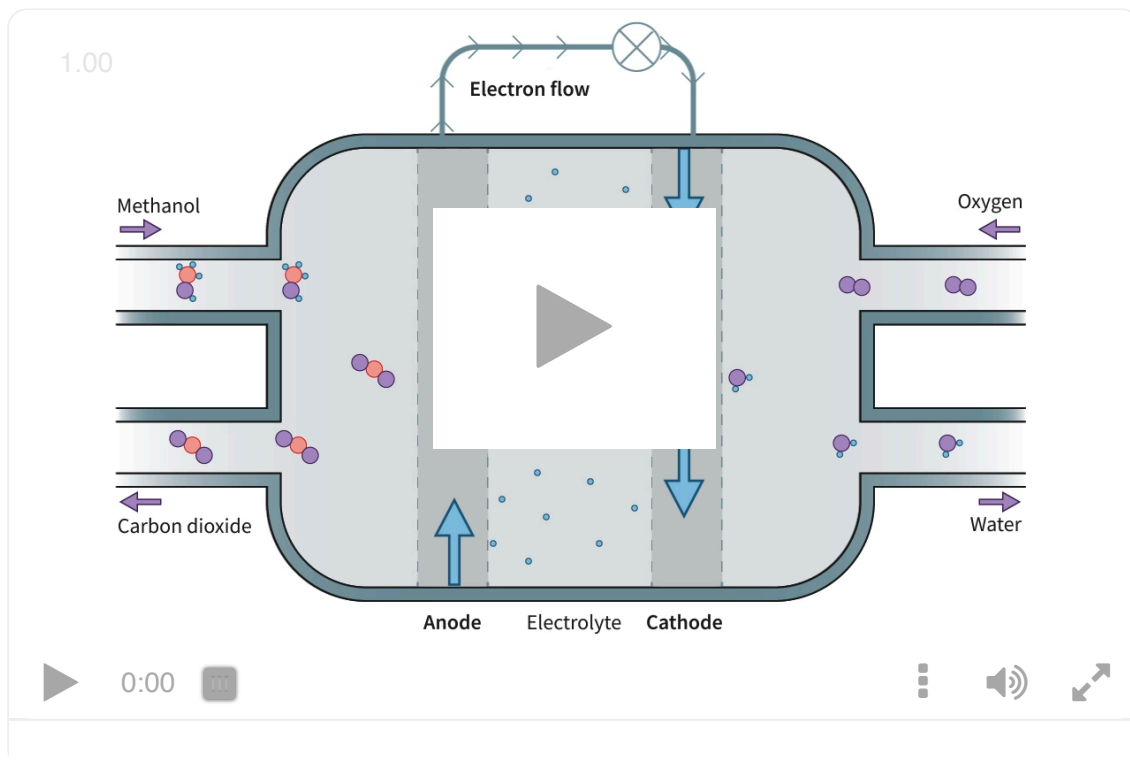


Adding these two half-reactions together produces the following net-reaction, which you can see is the same as the combustion of methanol with reduced coefficients. You may have noticed that combustion reactions are sometimes balanced using fractions as coefficients in order to make sure that the combustion reaction is written per mole of fuel combusted. This allows for a better comparison of energy produced per mole of fuel. In addition, the balanced overall reaction must be equal to the sum of the two half-reactions, including the correct states of matter and molar coefficients.



When electrons are lost by an element the element is said to have undergone oxidation. When electrons are gained by an element the element is said to have been reduced. These concepts will be explored in more detail in [section R3.2.1](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/>).

Use **Interactive 1** to explore how a methanol fuel cell functions.



Interactive 1. Functioning of a Methanol Fuel Cell.

 More information for interactive 1

This diagram shows how a methanol fuel cell generates electricity by converting chemical energy into electrical energy through a redox reaction.

On the left side, methanol (CH_3OH) and water (H_2O) enter the fuel cell through an inlet. A chemical reaction occurs here where methanol is oxidised, releasing carbon dioxide (CO_2), hydrogen ions (H^+), and electrons (e^-).

The hydrogen ions move through a special membrane in the centre of the cell, while the electrons travel through an external wire, creating an electric current that powers the bulb or the device.

On the right side, oxygen gas (O_2) enters through another inlet. Here, the oxygen combines with the hydrogen ions and electrons, forming water (H_2O) in a reduction reaction. The flow of electric current is indicated by the continuous flow of an upward arrow labelled anode and the continuous flow of a downward arrow labelled cathode. On the left, water is emitted out through the outlet. This setup is an efficient way to produce electricity, with only carbon dioxide and water as byproducts.

This interactive explains how a methanol fuel cell works using the concepts of oxidation, reduction, and ion flow. Identify the inputs (methanol, water, oxygen) and outputs (carbon dioxide, water, electrons) of the reaction.

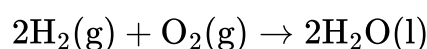
It is interesting to note that fuel cells are supplied with a continuous flow of reactant molecules, which are pumped into the cell. Unlike stand-alone batteries, fuel cells do not need to be charged over long periods. Methanol fuel cells can be refilled from a liquid methanol reservoir, which makes them convenient to use.

Hydrogen fuel cells

Since the advent of space exploration NASA and other agencies have been using hydrogen fuel cells as a portable method of powering the electrical systems on board spacecraft. A similar process occurs in hydrogen fuel cells compared to methanol fuel cells. In this case, the spontaneous and exothermic reaction between hydrogen gas and oxygen gas to form water is harnessed. This reaction releases approximately 286 kJ of energy per mole of hydrogen gas reacted.

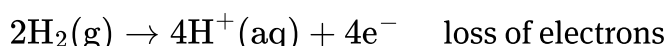
Let us examine the fuel cell under acidic conditions.

Where the net reaction:

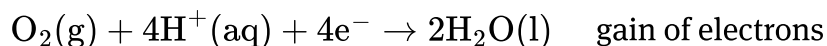


is divided into the following two half-reactions.

Half reaction:



Half reaction:



Making connections

As seen in [section S3.1.6 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-states-id-44696/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-states-id-44696/). At the anode, the hydrogen atoms in hydrogen gas have an oxidation state of zero, yet as hydrogen ions, they have an oxidation state of +1. This change in oxidation state from 0 to +1 involves the loss of electrons. Loss of electrons is said to occur at the anode.

On the other hand, at the cathode, the oxygen atoms in oxygen gas have an oxidation state of zero, yet as oxygen in water molecules, they have an oxidation state of -2 . This change in oxidation state from 0 to -2 involves the gain of electrons. The gain of electrons is said to occur at the cathode.

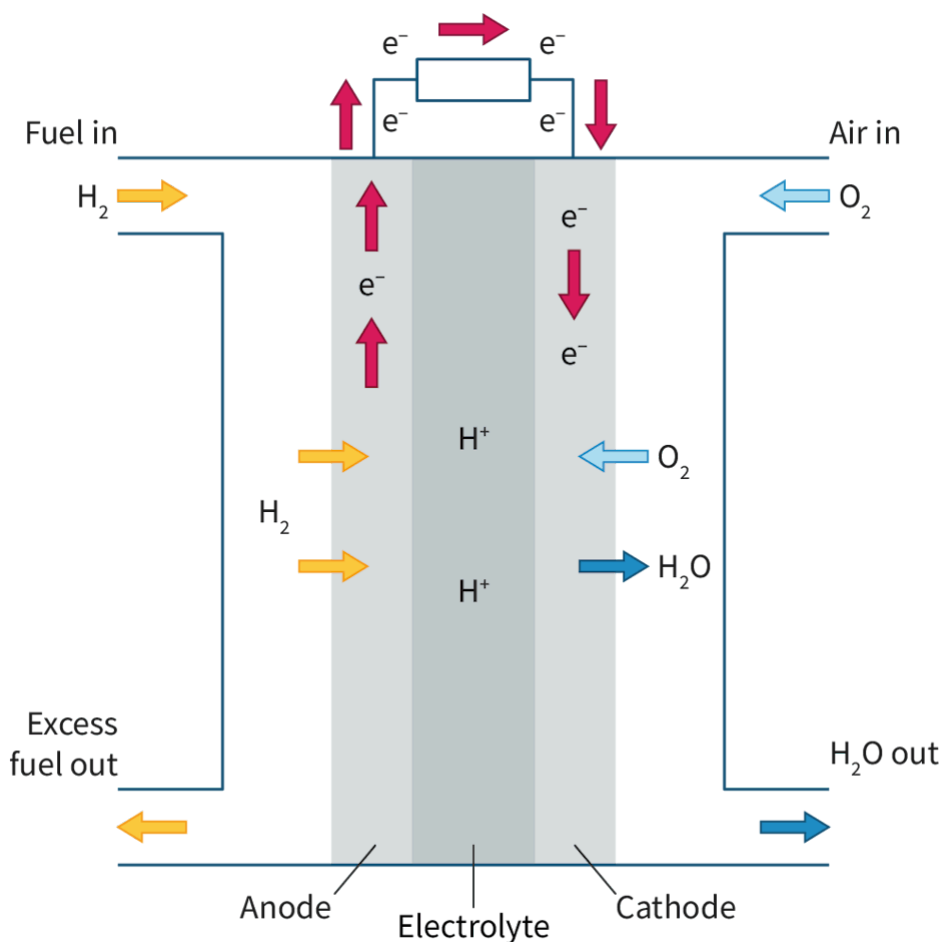


Figure 2. A hydrogen fuel cell.

 More information for figure 2

The diagram illustrates a hydrogen fuel cell. It comprises three main sections: anode, cathode, and the central electrolyte area where the reaction occurs. On the left, labeled "Fuel in," hydrogen gas enters the anode. On the right, "Air in" is labeled, showing oxygen entering the cathode. Water is labeled as exiting at the bottom, "H₂O out." Inside the cell, arrows indicate directional flow; upward at the anode and downward at the cathode. A symbol for electron flow moves from the anode to the cathode above the main cell body, indicating the generation of electricity.

[Generated by AI]

Though hydrogen fuel cells produce only water as products, it is always important to consider the overall reaction when assessing its environmental impact. It is energetically very difficult to obtain hydrogen gas, as most hydrogen gas on Earth is trapped in the form of water. To free the hydrogen in water it must go through a process known as electrolysis, which requires a tremendous amount of energy.

Hydrogen gas is also produced by a chemical process known as coal gasification, where coal is heated in the presence of steam to produce hydrogen gas and carbon monoxide gas. This method of producing hydrogen gas yields the same atmospheric pollutant as the incomplete combustion of fossil fuels and is therefore not environmentally sustainable. The process also requires a considerable amount of energy as the reaction takes place at high temperatures and pressures.

Activity

- **IB learner profile attribute:**
 - Knowledgeable
 - Communicator
- **Approaches to learning:**
 - Communication skills — Applying interpretive techniques to different forms of media
 - Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Individual activity

Instructions

1. Study this interactive:
<https://sepup.lawrencehallofscience.org/sepup-fuel-cell-simulation/> 
(<https://sepup.lawrencehallofscience.org/sepup-fuel-cell-simulation/>)
2. Using the interactive:
 - (a) State the half-reaction that would occur at the anode in a hydrogen fuel cell.
 - (b) State the half-reaction that would occur at the cathode in a hydrogen fuel cell.
 - (c) List the products in the overall reaction in a hydrogen fuel cell.
 - (d) Describe if the overall reaction absorbs or releases energy.

(e) Compare the enthalpy of the products with the enthalpy of reactants in the overall hydrogen fuel cell reaction.